DATA DICTIONARY

Amazon India: A Decade of Sales Analytics

11 Years of E-Commerce Data · 1.1 Million Transactions · PostgreSQL Star Schema

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1. Project Overview

Attribute	Detail
Project Name	Amazon India: A Decade of Sales Analytics
Data Period	2015 – 2025 (11 years)
Raw Transactions	1,127,609 rows across 11 annual CSV files
After Cleaning	1,122,000 rows (99.5% retention rate)
Products	2,004 unique SKUs across 6 subcategories
Database	PostgreSQL 15+ — Star schema with 4 tables
ETL Pipeline	Python 3.11 · SQLAlchemy · Pandera validation
Subcategories	Smartphones, Laptops, Tablets, Smart Watch, Audio, TV & Entertainment
Geographic Scope	30+ Indian cities — Metro / Tier1 / Tier2 / Rural

2. Dataset 1 — Sales Transactions

Source: **data/raw/amazon_india_{year}.csv** (11 files, 2015–2025) Cleaned output: **data/processed/cleaned_df_sales.csv**

2.1 Column Reference

Column	Data Type	Description	Example
transaction_id	VARCHAR	Unique transaction identifier	TXN_2019_000001
order_date	DATE	Order date — standardised to YYYY-MM-DD	2019-10-15
order_month	SMALLINT	Month number extracted from order_date	10
order_year	SMALLINT	Year extracted from order_date	2019
order_quarter	SMALLINT	Quarter number (1–4)	4
customer_id	VARCHAR	Unique customer identifier	CUST_000123
customer_city	VARCHAR	Standardised city name (Bangalore normalised to Bengaluru)	Bengaluru
customer_state	VARCHAR	Indian state	Karnataka
customer_tier	VARCHAR	City tier classification: Metro / Tier1 / Tier2 / Rural	Metro
customer_spending_tier	VARCHAR	Spending behaviour segment: High / Medium / Low	High
customer_age_group	VARCHAR	Age bracket: 18-25 / 26-35 / 36-45 / 46-55 / 55+	26-35
is_prime_member	BOOLEAN	Prime membership flag. Cleaned from Yes/No/1/0/Y/N	True
product_id	VARCHAR	Unique product identifier	PROD_000456
product_name	VARCHAR	Product display name	Samsung Galaxy S21
category	VARCHAR	Top-level category — 'Electronics' for all rows in this dataset	Electronics

Column	Data Type	Description	Example
subcategory	VARCHAR	Product subcategory — 6 distinct values; use this for analysis	Smartphones
brand	VARCHAR	Brand name (100+ brands)	Samsung
is_prime_eligible	BOOLEAN	Whether product qualifies for Prime free delivery	True
product_weight_kg	NUMERIC	Product weight in kilograms	0.18
product_rating	NUMERIC	Product catalogue rating 1.0–5.0	4.2
original_price_inr	NUMERIC	List price before discount. Cleaned: removed Rs symbols, commas, text	75000.00
discount_percent	NUMERIC	Discount applied as percentage (0–100)	15.0
discounted_price_inr	NUMERIC	Price after discount applied	63750.00
quantity	SMALLINT	Units ordered (typically 1–3 units)	1
subtotal_inr	NUMERIC	discounted_price_inr multiplied by quantity	63750.00
delivery_charges	NUMERIC	Delivery fee in INR. Zero for Prime-eligible orders	200.00
final_amount_inr	NUMERIC	Total amount paid: subtotal plus delivery charges	63950.00
payment_method	VARCHAR	Standardised payment method. Cleaned: UPI/PhonePe/GooglePay all mapped to UPI	UPI
payment_category	VARCHAR	Grouped payment type: Digital / Card / Cash	Digital
delivery_days	NUMERIC	Days from order placement to delivery. Cleaned from text values like 'Same Day'	3.0
delivery_type	VARCHAR	Delivery speed tier: Same Day / Express / Standard	Standard
is_festival_sale	BOOLEAN	Festival period order flag. Cleaned from inconsistent True/False/1/0	False
festival_name	VARCHAR	Festival name when is_festival_sale is True; NULL otherwise	Diwali
sale_type	VARCHAR	Sale classification: Festival / Regular	Regular
return_status	VARCHAR	Order outcome: Delivered / Returned / Cancelled	Delivered
customer_rating	NUMERIC	Post-purchase customer rating 1.0–5.0. Imputed if missing	4.0
rating_missing	SMALLINT	Flag: 1 if customer_rating was imputed; 0 if original	0

2.2 Data Quality Issues Fixed (10 Cleaning Challenges)

#	Issue	Method Applied	Rows Affected
1	Mixed date formats: DD/MM/YYYY, DD-MM-YY, YYYY-MM-DD, and invalid entries	Multi-format parse with fallback; invalid dates → sentinel 1900-01-01	~4,757 rows
2	Price column: Rs symbols, Indian comma separators, text like 'Price on Request'	Regex strip all non-numeric; text entries → NaN → median imputation per subcategory	~2,898 rows

#	Issue	Method Applied	Rows Affected
3	Ratings in multiple formats: '4 stars', '3/5', '2.5/5.0', missing values	Regex numeric extraction; fraction normalisation to 1.0–5.0 scale	~15,336 rows
4	Inconsistent city names: Bangalore / Bengaluru, spelling errors, case variations	Canonical mapping dictionary with 30+ aliases; lowercase normalisation	~222 rows
5	Boolean columns with mixed: True/False, Yes/No, 1/0, Y/N, and nulls	Explicit mapping table for all variants → Python bool	All boolean columns
6	Category name variations: Electronics / ELECTRONICS / Electronic & Accessories	Lowercase + canonical string mapping to 'Electronics'	~0 after standardise
7	Delivery days: negative values, text like 'Same Day' or '1-2 days', values > 30	Text parsing → numeric; negatives → 0; IQR-based cap at 95th percentile per tier	~110 rows
8	Duplicate transactions: same customer, product, date, and amount	Keep first occurrence; bulk orders identified separately via quantity > 1	~244 removed
9	Price outliers: prices 100x expected value due to decimal point entry errors	IQR method per subcategory; decimal shift correction applied	~1,457 fixed
10	Payment method naming: UPI / PhonePe / GooglePay, Credit Card / CC / CREDIT_CARD	Canonical mapping to 7 standard categories: UPI, COD, Credit Card, Debit Card, Net Banking, Wallet, BNPL	All payment rows

3. Dataset 2 — Product Catalog

Source: **data/raw/amazon_india_products_catalog.csv** Output: **data/processed/cleaned_products.csv** 2,004 rows x 11 columns. Joined to fact_transactions on product_id.

Column	Data Type	Description	Example
product_id	VARCHAR PK	Unique product identifier — foreign key in fact table	PROD_000456
product_name	VARCHAR	Full product display name	Samsung Galaxy S21
category	VARCHAR	Top-level category ('Electronics' for all rows)	Electronics
subcategory	VARCHAR	Subcategory — 6 distinct values; use for all analysis	Smartphones
brand	VARCHAR	Brand name (100+ distinct brands in dataset)	Samsung
base_price_2015	NUMERIC	Original launch price in 2015 (INR) — inflation baseline	45000.00
weight_kg	NUMERIC	Product physical weight in kilograms	0.18
rating	NUMERIC	Catalogue aggregate rating averaged across all reviews	4.3
is_prime_eligible	BOOLEAN	Whether product qualifies for Amazon Prime free delivery	True
launch_year	SMALLINT	Year product was first listed on Amazon India	2021
model	VARCHAR	Model or variant identifier string	SM-G991B

4. PostgreSQL Star Schema

After ETL, all data is loaded into a PostgreSQL star schema. Schema defined in **scripts/init_db.sql** and created automatically on first run. Upsert strategy (INSERT ... ON CONFLICT DO UPDATE) makes ETL re-runnable safely.

Table	Type	Full Dataset Rows	Purpose
fact_transactions	Fact	~1,122,000	One row per transaction — all numeric measures
dim_customers	Dimension	~354,000	Customer master: city, tier, age group, Prime flag
dim_products	Dimension	2,004	Product master: category, brand, launch year
dim_time	Dimension	~4,000	Date dimension: year, month, quarter, festival flag

4.1 fact_transactions — Central Fact Table

One row per transaction. Foreign keys to all 3 dimension tables. Upserted on transaction_id.

Column	Type	Constraint / Notes
transaction_id	VARCHAR	PRIMARY KEY
order_date	DATE	Indexed — primary date filter
order_year	SMALLINT	Indexed — used in all annual aggregations
order_month	SMALLINT	
order_quarter	SMALLINT	
customer_id	VARCHAR	FK — dim_customers.customer_id
product_id	VARCHAR	FK — dim_products.product_id
original_price_inr	NUMERIC	NULL allowed for products without list price
discount_percent	NUMERIC	0 to 100
discounted_price_inr	NUMERIC	
final_amount_inr	NUMERIC	Indexed — used in all revenue aggregations
quantity	SMALLINT	Typically 1–3 units
payment_method	VARCHAR	7 standard values after cleaning
delivery_days	NUMERIC	NULL if order not yet delivered
delivery_type	VARCHAR	Same Day / Express / Standard
is_festival_sale	BOOLEAN	
festival_name	VARCHAR	NULL when is_festival_sale = False
product_rating	NUMERIC	1.0 to 5.0
customer_rating	NUMERIC	1.0 to 5.0; NULL if customer has not rated
return_status	VARCHAR	Delivered / Returned / Cancelled

4.2 dim_customers — Customer Dimension

One row per unique customer. Populated from distinct customer_ids in fact_transactions.

Column	Type	Constraint / Notes
customer_id	VARCHAR	PRIMARY KEY
customer_city	VARCHAR	Standardised city name
customer_state	VARCHAR	Indian state
customer_tier	VARCHAR	Metro / Tier1 / Tier2 / Rural
customer_spending_tier	VARCHAR	High / Medium / Low
customer_age_group	VARCHAR	18-25 / 26-35 / 36-45 / 46-55 / 55+
is_prime_member	BOOLEAN	

4.3 dim_products — Product Dimension

One row per unique product. Sourced from the product catalogue CSV.

Column	Type	Constraint / Notes
product_id	VARCHAR	PRIMARY KEY
product_name	VARCHAR	
category	VARCHAR	'Electronics' for all rows in this dataset
subcategory	VARCHAR	6 distinct values — always use subcategory for analysis
brand	VARCHAR	100+ brands
launch_year	SMALLINT	Used for product lifecycle analysis
is_prime_eligible	BOOLEAN	

4.4 dim_time — Date Dimension

Date dimension for time-based joins. One row per calendar date in the dataset range.

Column	Type	Constraint / Notes
date	DATE	PRIMARY KEY
year	SMALLINT	
month	SMALLINT	1 to 12
quarter	SMALLINT	1 to 4
is_festival_period	BOOLEAN	True for known festival windows: Diwali, Prime Day, Great Indian Festival, etc.

5. Key Business Metrics Reference

Key figures from the full 1.1M row dataset for quick reference during analysis and presentation.

Metric	Value	Context
Total raw transactions	1,127,609	Across 11 annual CSV files
After ETL cleaning	1,122,000	99.5% retention rate
Unique customers	~354,000	Long-term customer base
Unique products (SKUs)	2,004	Electronics catalog
Dataset period	2015 to 2025	11 complete years
Subcategories (6 total)	Smartphones, Laptops, Tablets, Smart Watch, Audio, TV	Use subcategory not category
States covered	15+ Indian states	
Cities covered	30+ cities	Metro to Rural
Smartphone revenue share	73.2%	India is mobile-first
Peak revenue year	2020	COVID e-commerce surge
10-Year Revenue CAGR	6.2%	Healthy mature-market growth
Prime vs Non-Prime AOV	Rs 78,127 vs Rs 62,296	+25% for Prime members
Average Customer CLV (mean)	Rs 74,100	Right-skewed distribution
Average Customer CLV (median)	Rs 47,900	More representative typical value
Festival AOV vs Regular AOV	Rs 47,400 vs Rs 77,700	Festival is LOWER — discount effect
Highest return rate category	Audio at 8.1%	Subjective sound quality
Lowest return rate category	TV & Entertainment 5.1%	Screen size meets expectations
Optimal discount range	20 to 30%	Highest revenue per INR discounted
Average delivery time	3.3 to 3.5 days all tiers	Rural only 0.2 days slower than Metro
Top 36% products	Generate 80% revenue	Pareto concentration — 720 SKUs drive the business
UPI payment share 2025	Largest and growing	Displaced COD since 2016 demonetisation
BNPL return rate	7.8% vs 6.7% UPI	Deferred payment reduces commitment